

Multi-UAV Source Seeking and Optimization

BTP Project

*Dikshant (2022102038) ,
Dr. Harikumar Kandath*

September 15, 2025

Continuing BTP-1 work....Recap

What was done in BTP-1:

- Last sem the work focused on understanding different PSOs.
- Implementing [2].
- Doing exploring based source seeking for APSO, ARPSO, SPSO.

What is plan for BTP-2:

- This sem target is to try exploitation strategies in source seeking algos and observe behaviours of different PSOs
- To identify which PSO or combination of different PSOs work best in which scenario.

PSO is the technique used to find an approximate solution for maximization or minimization problems.

- PSO is initialized with a group of random particles (solutions) and then searches for the optimal solution by updating generations.
- Particles move through the solution space, and are evaluated according to some fitness criterion after each time step. In every iteration, each particle is updated by following two "best values".
- The first one is the best solution (fitness) it has achieved so far (the fitness value is also sorted). This value is called P_{best} .
- Another best value that is tracked by the PSO is the best value obtained so far by any particle in the population. This second-best value is a **global best** and called g_{best} .

Objective

Source Seeking:

- Understand [2] and reproduce its results.

Optimization:

- Using CEC2022 benchmarks [1] for optimization of [2] PSO algorithms.
- To develop and evaluate hybrid Particle Swarm Optimization approaches that combine the strengths of different PSO variants
- To improve optimization performance on complex benchmark functions through strategic algorithm hybridization
- To analyze how different hybridization strategies affect convergence behavior and solution quality
- To identify which hybrid approaches are most effective for specific types of optimization problems

APSO (Acceleration based PSO)

- Uses acceleration as the primary update mechanism (second-order dynamics)
- Maintains momentum through previous movement history
- Produces smoother trajectories with more consistent direction

Mathematical Formulation:

- **acc update:** $a_i(t+1) = \omega_1 a_i(t) + c_1 r_1 (p_i - x_i) + c_2 r_2 (g - x_i)$
- **Velocity update:** $v_i(t+1) = \omega_2 v_i(t) + a_i(t+1) T$
- **Position update:** $x_i(t+1) = x_i(t) + v_i(t+1) T$

SPSO (Standard PSO)

- Direct velocity-based update mechanism (first-order dynamics)
- Balanced exploration-exploitation trade-off
- Widely used baseline algorithm in swarm intelligence

Mathematical Formulation:

- **Velocity update:** $v_i(t+1) = \omega_1 v_i(t) + c_1 r_1 (p_i - x_i) + c_2 r_2 (g - x_i)$
- **Position update:** $x_i(t+1) = x_i(t) + v_i(t+1)$

ARPSO (Adaptive Random PSO)

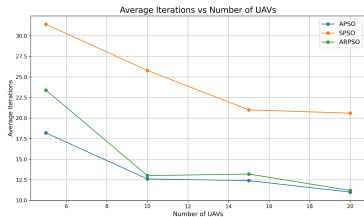
- Incorporates adaptive parameters and randomness
- Dynamically adjusts inertia weight based on swarm state
- Includes additional random attractive positions for exploration

Mathematical Formulation:

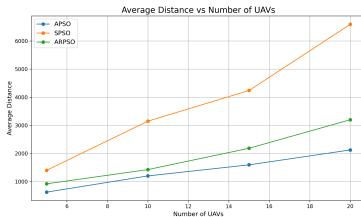
- **Adaptive inertia:** $\omega_i(t+1) = f(\text{evolutionary speed}, \text{aggregation degree})$
- **Velocity update:** $v_i(t+1) = \omega_i v_i(t) + c_1 r_1 (p_i - x_i) + c_2 r_2 (g - x_i) + c_3 r_3 (a_i - x_i)$
- here a_i is a random attractive position
- **Position update:** $x_i(t+1) = x_i(t) + v_i(t+1)$

NOTE: For code all the values were taken from the paper [2].

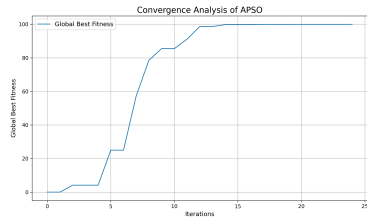
Source Seeking without noise



(a) iterations vs No. of UAV's



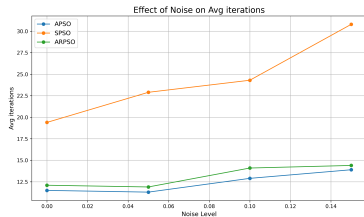
(b) Distance vs No. of UAV's



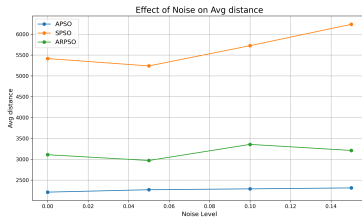
(c) Convergence analysis

Figure: Source Seeking without Noise

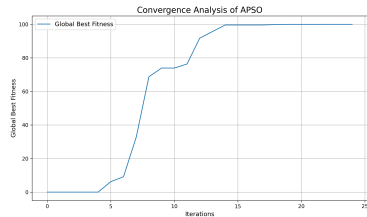
Source Seeking with noise



(a) iterations vs Noise



(b) Distance vs Noise



(c) Convergence analysis

Figure: Source Seeking with Noise for 30 UAV's

Source Seeking Conclusions

- APSO consistently outperforms both SPSO and ARPSO across all noise levels.
 - ▶ APSO requires fewer iterations
 - ▶ APSO travels less distance
- APSO shows the best scaling as UAV count increases, iterations actually decrease and SPSO shows the worst scaling in terms of distance traveled.
- All algorithms show degraded performance as noise increases
 - ▶ The number of iterations increases with noise level
 - ▶ The distance traveled also generally increases with noise level
- APSO shows the best robustness to noise
 - ▶ The slope of the APSO line is less steep than SPSO, indicating better resilience to noise
 - ▶ Even at 0.15 noise level, APSO maintains good performance
- APSO provides smoother convergence under noise. As its acceleration based approach allows it to maintain momentum through noisy measurements, while the standard PSO gets easily misled by noise.

CEC'22 Benchmark Suit

What are CEC Benchmarks?

- Standard test suite developed by IEEE Congress on Evolutionary Computation
- CEC 2022 includes diverse optimization problems with known properties
- Designed to rigorously evaluate and compare optimization algorithms

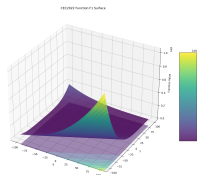
Functions Implemented:

- **F1 (Zakharov):** Continuous unimodal function with interdependent variables
- **F2 (Rosenbrock):** Continuous function with narrow curved valley
- **F3 (Schaffer):** Highly multimodal function with many local optima
- **F4 (Step Rastrigin):** Discontinuous multimodal function with plateaus
- **F5 (Levy):** Multimodal function with many local minima

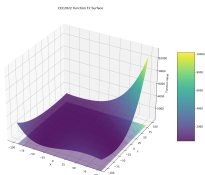
Why these functions??

- Represent diverse optimization challenges
- **F1-F2:** Test convergence precision and speed
- **F3-F5:** Test ability to escape local optima
- Combination reveals which hybrid approaches excel in different scenarios

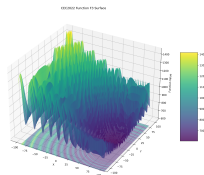
CEC'22 Visualization



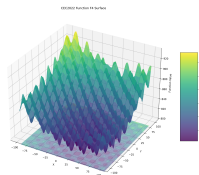
F1



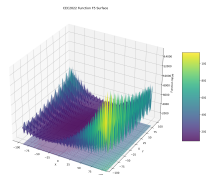
F2



F3



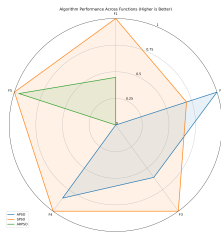
F4



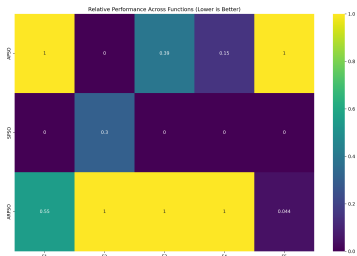
F5

Function surfaces for CEC benchmark functions

APSO Optimization Visualization



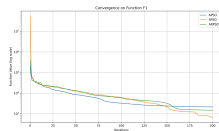
PSO's Performance
across functions



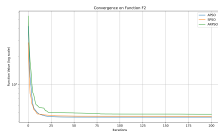
Relative Performance Across function

Performance Matrix

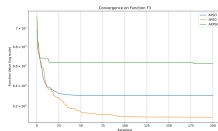
For 2D,3D visualization of APSO optimization on Functions, visit: [@pingu-73/vigilant-spork/..gifs](https://github.com/pingu-73/vigilant-spork)



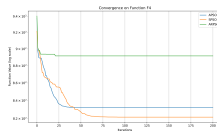
F1 convergence



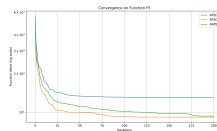
F2 convergence



F3 convergence



F4 convergence



F5 convergence

Convergence Plots

Observations

- **SPSO performs best overall:** The radar chart and numerical results show SPSO generally finding better solutions (lower values) across most functions, particularly on F1, F3, F4, and F5.
- **APSO shows mixed performance:** It performs well on some functions (F2, F4) but struggles with others (F1, F5), showing higher variability (larger standard deviations).
- **ARPSO consistently underperforms:** It generally finds worse solutions than the other algorithms, especially on F3 and F4, which matches its design as prioritizing exploration over exploitation.

- **F1 (Zakharov Function)**

- ▶ SPSO performs best, APSO shows high variability
- ▶ The convergence plot shows SPSO eventually finding better solutions after 150+ iterations
- ▶ **Zakharov relatively smooth but has interdependent variables. APSO's momentum can sometimes overshoot, explaining its variability, while SPSO's balanced approach works well.**

- **APSO:** faster initial convergence (F2, F4) but plateaus earlier. This matches its acceleration based design , momentum helps it make quick progress but make fine adjustments difficult.
- **SPSO:** steady, consistent improvement and achieves the best final solutions (F1, F3, F5). Its balanced approach allows it to both explore broadly and refine solutions effectively.
- **ARPSO:** plateaus early (F3, F4) but occasionally shows late improvements (F1). The added randomness helps it escape local optima but can prevent focused exploitation.

Summary

- Each PSO's performance is strongly influenced by how well its movement dynamics match the characteristics of the fitness landscape.
- APSO's acceleration-based approach provides smoother trajectories but at the cost of adaptability in complex landscapes.
- SPSO maintains the best balance for these benchmark functions, while APSO tends toward exploitation and ARPSO toward exploration.
- The PSO shows complementary strengths that could potentially be combined in hybrid approaches using APSO's momentum for initial exploration and SPSO's adaptability for final refinement.

Exploitation/Hybrid Approach

Hybrid approaches used:

● Particle Hybrid

- ▶ Different particles within the same swarm use different update rules
- ▶ The swarm is divided into 3 groups of 10 particles each
- ▶ Group1: APSO, Group2: SPSO, Group3: ARPSO
- ▶ All particles share information about global best position

● Sequential Hybrid

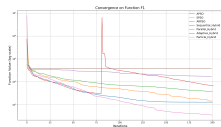
- ▶ Different algorithms used in sequence, with one algorithm taking over where the previous left off.
- ▶ The swarm is divided into 3 groups of 10 particles each
- ▶ Phase 1: APSO runs for 40% of total iterations
- ▶ Phase 2: SPSO is initialized around APSO's final solution and runs for 60% of iterations
- ▶ The best solution from APSO is used as a starting point for SPSO

● **Parallel Hybrid**

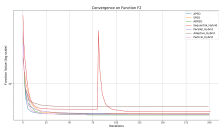
- ▶ Multiple algorithms run simultaneously on separate sub-swarms and share information.
- ▶ Three separate swarms run in parallel (10 particles each)
- ▶ Swarm1: APSO, Swarm2: SPSO, Swarm3: ARPSO
- ▶ After each iteration, the best solution across all swarms is shared with all algorithms

● **Adaptive Hybrid**

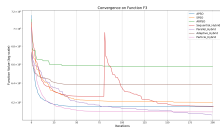
- ▶ Dynamically switches between algorithms based on their recent performance.
- ▶ Maintains three algorithm instances (APSO, SPSO, ARPSO)
- ▶ Starts with APSO, Every 10 iterations, evaluates which algorithm has made the most improvement
- ▶ Switches to the best performing algorithm for the next 10 iterations, all algorithms share the same global best information



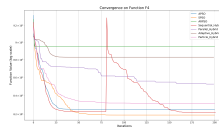
F1 convergence



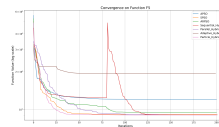
F2 convergence



F3 convergence



F4 convergence



F5 convergence

Convergence Plots

Observations

- **Anomaly in Sequential_Hybrid convergence plots:** sudden spike around iteration 75 is due to the transition between APSO and SPSO phases. When SPSO takes over, it's initialized around the APSO solution but with some random perturbation, which temporarily worsen the fitness before improving it.
- Particle_Hybrid performs well overall (avg rank 2.20, 1 win)
- Parallel_Hybrid is also strong (avg rank 3.20, 2 wins)
- APSO and SPSO tie for third (avg rank 3.60)
- Sequential_Hybrid is fifth (avg rank 4.00)
- ARPSO is sixth (avg rank 4.80, 1 win)
- Adaptive_Hybrid performs worst (avg rank 6.60)
- Particle_Hybrid and Parallel_Hybrid show the most promise overall, confirming that combining different PSO variants can improve performance.

Ramining Work

For stong conclusive results

- Reduce or remove random perturbation from SPSO in the Sequential_hybrid function
- Do more runs (10-30) as the current 5 runs are not enough for statistical significance.

Codebase: github.com/pingu-73/vigilant-spork & github.com/pingu-73/source-seeking/

References



Wenjian Luo, Xin Lin, Changhe Li, Shengxiang Yang, and Yuhui Shi.

Benchmark functions for cec 2022 competition on seeking multiple optima in dynamic environments, 2022.



Adithya Shankar, Himanshu, Harikumar Kandath, and J. Senthilnath.

Acceleration-based pso for multi-uav source-seeking.

In *IECON 2023- 49th Annual Conference of the IEEE Industrial Electronics Society*, pages 1–6, 2023.